

Liang (Twist) Shan



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🏠 <https://twist-shan.github.io/>

RESEARCH INTEREST



- Statistical foundations, optimization, and mechanistic interpretability of **Modern Machine Learning**, with a focus on deep learning, reinforcement learning, and generative AI, including language models, diffusion models and AI agents.

EDUCATION



Peking University

Beijing, China

- *B.S. in Statistics, School of Mathematical Sciences* Jun. 2024–Jul. 2028 (*Expected*)
Second Major B.A. in Economics, National School of Development Sep. 2025–Jul. 2028 (*Expected*)
 - **Statistics Courses:** Probability Theory, Mathematical Statistics, Applied Stochastic Processes, Statistical Thinking, Stochastic Analysis and Applications.
 - **Computer Science Courses:** Data Structure and Algorithm, Fundamentals of Artificial Intelligence,
 - **Finance and Economics Courses:** Microeconomics, Macroeconomics, Econometrics (Honor), Mathematical Economics, Introduction to Finance, Security Investment, Mathematical Methods in Finance, Modern Quantitative Trading Systems, Topics in Quantitative Finance.

University of Copenhagen

Copenhagen, Denmark

- *International Exchange Student, Department of SCIENCE* Sep. 2025–Jan. 2026
 - **Mathematics Courses:** Topics in Statistics (MCMC), Advanced Operation Research: Stochastic Programming.
 - **Computer Science Courses:** Probability Machine Learning

Peking University

Beijing, China

- *B.S. in Physics, School of Earth and Space Sciences* Sep. 2023–Jun. 2024

RESEARCH EXPERIENCES



Mechanistic Interpretability of Latent Reasoning in DLMs

University of California, Berkeley

- *Advisor: Prof. Song Mei, UC Berkeley* Aug. 2026 – Present
 - Designed tasks and interpretability pipeline to find out the reasoning mechanics in Diffusion Language Models between different architectures and decoding orders.

Pareto Frontier Model Identification from Pairwise Preferences

University of Pennsylvania

- *Advisor: Prof. Zhimei Ren, UPenn* May. 2026 – Sep. 2026
 - Developed an adaptive vector-Borda elimination algorithm for fixed-confidence Pareto frontier identification, with exact recovery and instance-dependent upper and minimax lower bounds on sample complexity.
 - Built regularized-MLE baselines, synthetic benchmarks, and robust preprocessing and replay pipelines for multi-objective evaluation on Real-world Arena data.

Mechanistic Interpretability of Chain-of-Thought Reasoning

University of Wisconsin–Madison

- *Advisor: Prof. Yiqiao Zhong, UW–Madison* Apr. 2026 – Sep. 2026
 - Developed controlled synthetic and realistic counting tasks, characterizing how chain-of-thought shifts computation from one-shot broad retrieval to iterative targeted retrieval and counter updates.
 - Built a causal interpretability pipeline using probing, hidden-state steering, position-local head ablation, and activation patching to localize retrieval heads and internal count representations.

SELECTED COURSE PROJECTS



LLM Applications: WeChat Content Agent

Peking University

- *Advisor: Prof. Shanghang Zhang, PKU* Apr. 2025 – Jun. 2025
 - Built a Dify-based LLM agent that automated content generation, responsive HTML/CSS typesetting, and draft publishing through Python and WeChat APIs.
 - Integrated RAG and web-retrieval tools to support source-grounded writing and dynamic content collection.

Empirical Asset Pricing and Portfolio Management

Peking University

Mar. 2025 – Jun. 2025

▪ Advisor: Prof. Hai Huang, PKU

- Analyzed Chinese A-share data and constructed dynamically rebalanced portfolios using mean–variance optimization and fundamental signals.
- Evaluated mutual-fund timing and stock-selection performance using the Treynor–Mazuy and Henriksson–Merton models.

[GitHub](#)

Quantitative Trading Backtesting Framework

Peking University

Sep. 2024 – Jan. 2025

▪ Advisor: Dr. Changhao Jiang, PKU

- Developed a modular Python backtesting framework that decoupled signal generation, order execution, and performance evaluation.
- Implemented and benchmarked momentum, mean-reversion, cross-sectional, and time-series strategies using vectorized data processing.

[GitHub](#)

SELECTED HONORS AND AWARDS



Yu Hao Dream Scholarship

Peking University

May. 2026

▪ Scholarship supporting international exchange program

SKILLS & OTHERS



- **Computer Skills:** Python (&PyTorch), R, C++ (Elementary), L^AT_EX
- **Languages:** Mandarin Chinese (Native), English (Fluent)
- **GRE General Test:** 325/340 (Verbal 155, Quantitative 170)